

As a Senior Frontend Architect, I have designed the architecture for the AI-Powered Hospital Management System focusing on **HIPAA compliance, high-performance data visualization, and role-based accessibility**.

1. UI Architecture

- **Framework:** Next.js 14+ (App Router) for SEO, server-side data fetching (essential for EHR speed), and edge-middleware security.
- **Language:** TypeScript (Strict mode).
- **Styling:** Tailwind CSS + Radix UI (accessible, unstyled primitives).
- **Communication:** GraphQL (Apollo Client) for flexible data querying of complex medical records; WebSockets (Socket.io) for real-time triage updates.

2. Pages

- **Public:** Landing, Auth (Login/MFA), Forgot Password.
- **Protected (Admin):** Global Dashboard, Staff Management, Billing Analytics.
- **Protected (Clinical):** Patient Triage Queue (Real-time), EHR Viewer, AI Diagnostic Workspace.
- **Protected (Patient):** My Records, Appointment Booking, Billing/Insurance portal.

3. Components

- **DataGrid/Table:** High-performance virtualization (TanStack Table) for large patient lists.
- **AI Insight Card:** A standardized component with "Human-in-the-Loop" verification and audit signatures.
- **Medical Input Controls:** Highly constrained inputs (e.g., date pickers, ICD-10 code lookups, unit-safe measurements).
- **Layout:** Responsive "Dashboard Shell" with a collapsible navigation sidebar and a persistent global notification header.

4. Routing

- **Protected Routes:** `/(dashboard)/[role]/...` protected via Next.js Middleware.
- **Sub-domains/Grouping:** Organized by user role to enforce security at the routing level.

- **Shallow Routing:** Used for patient records to allow switching between patients without re-fetching core layout components.

5. State Management

- **Server State:** TanStack Query (React Query) for caching EHR data and API responses.
- **Global Client State:** Zustand for lightweight UI state (sidebar toggle, current session user).
- **Real-time:** React Context + WebSockets for incoming triage requests.

6. Forms

- **Library:** React Hook Form + Zod (for schema-based validation).
- **Audit-enabled:** All form submissions are wrapped in a decorator that adds a mandatory "Reason for Change" and "User Fingerprint" for HIPAA compliance.

7. Authentication Flow

- **Method:** OAuth 2.0 + OpenID Connect (e.g., Auth0 or Keycloak).
- **Security:** Enforced MFA for all medical staff.
- **RBAC/ABAC:** Role-Based Access Control handled via JWT claims; UI components conditionally rendered based on `user.permissions` array.

8. Dashboard Layout

- **Layout Structure:**
 - * **Sidebar:** Persistent nav, Role-specific links.
 - * **Top Bar:** Search (Global patient search), Notifications, AI Status Indicator (System health).
 - * **Main Content Area:** Tabbed view for multitasking (e.g., viewing a lab result while updating a prescription).
 - * **Right Rail (AI Assistant):** A persistent chat/side-panel that provides contextual AI insights for the current page.

9. Folder Structure (Feature-based)

```
/src
  /app           # Next.js App Router
  /components
    /ui          # Radix primitives
    /shared      # Common UI (Buttons, Inputs)
    /dashboard   # Layout components
```

```
    /ehr          # Specialized medical widgets
    /hooks        # Custom hooks (e.g., usePatientTriage, useAudit)
    /lib          # Utils, API clients (Apollo, Zod)
    /store        # Zustand stores
    /types        # TypeScript interfaces/types
    /services     # API calls to backend/AI microservices
```

10. Recommended UI Libraries

- **Charts:** ``Recharts`` or ``Visx`` (for patient vitals and predictive analytics).
- **Icons:** ``Lucide React`` (Clean, professional, accessible).
- **Notifications:** ``Sonner`` (Minimalist, accessible toasts).
- **Forms/Validation:** ``Zod`` (Runtime validation).
- **Accessibility:** ``React-Aria`` (To ensure clinical interfaces are usable by everyone).

Architect's Note on "AI Hallucination" & Risk

To meet the "Human-in-the-loop" requirement, the UI will implement a **"Confidence Score Overlay"** on all AI suggestions. If the AI confidence is below 85%, the UI will force a secondary confirmation step, and the "Accept" button will be disabled until the clinician explicitly reviews the suggestion, creating an immutable ``SignedAction`` record in the system.